EXPLORATORY DATA ANALYSIS

Used Cars Market in India

Cleaning, exploring and visualising 15,411 CarDekho listings

Project title	Exploratory Data Analysis on a Cars Dataset
Student name	[Your Name]
Batch	[Your Batch]
Date	31 October 2026
Dataset	CarDekho Used Car Dataset (15,411 listings, 13 original columns)
Dataset source	CarDekho used-car listings (cardekho.com), obtained from the public GitHub repository manishkr1754/CarDekho_Used_Car_Price_Prediction, file notebooks/data/cardekho_dataset.csv (https://github.com/manishkr1754/CarDekho_Used_Car_Price_Prediction)
Tools	Python 3, pandas, NumPy, Matplotlib, Seaborn

1. Objective

The purpose of this project is to carry out a complete Exploratory Data Analysis (EDA) of a used-car dataset: to understand its structure, improve its quality, identify patterns and outliers, and communicate the results through clear visualisations. No machine-learning model is built. The analysis is guided by the following questions:

- Which brands and fuel types are the most common in the market?
- How are used-car selling prices distributed?
- How does vehicle age relate to price, and does distance driven (km) influence price?
- Do automatic and manual cars differ in price?
- Which brands have the highest average price?
- Which numerical features are strongly correlated with each other and with price?
- Where are the important outliers and unusual records?

2. Dataset Overview

The dataset contains **15,411 rows and 14 columns** (13 data columns plus an index column that was saved by mistake when the file was exported). Each row is one used car listed for sale on CarDekho. **Numerical variables**: vehicle_age, km_driven, mileage, engine, max_power, seats, selling_price. **Categorical variables**: car_name, brand, model, seller_type, fuel_type, transmission_type.

Column	Type	Description
car_name / brand / model	text	Full car name, manufacturer and model name
vehicle_age	int	Age of the car in years
km_driven	int	Total distance driven (km)
seller_type	text	Dealer, Individual or Trustmark Dealer
fuel_type	text	Petrol, Diesel, CNG, LPG or Electric
transmission_type	text	Manual or Automatic
mileage	float	Fuel efficiency (kmpl)
engine	int	Engine capacity (cc)
max_power	float	Maximum power (bhp)
seats	int	Number of seats
selling_price	int	Asking price in INR (target variable of interest)

Sample records (first 5 rows after cleaning):

car_name	age	km_driven	fuel	transmission	mileage	engine	max_power	price (INR)
Maruti Alto	9	120,000	Petrol	Manual	19.7	796	46.3	120,000
Hyundai Grand	5	20,000	Petrol	Manual	18.9	1197	82.0	550,000
Hyundai i20	11	60,000	Petrol	Manual	17.0	1197	80.0	215,000
Maruti Alto	9	37,000	Petrol	Manual	20.92	998	67.1	226,000
Ford Ecosport	6	30,000	Diesel	Manual	22.77	1498	98.59	570,000

3. Data Quality and Cleaning

Checks performed. No missing values were found in any column. Checking for exact duplicates (ignoring the index column) found **167 duplicate rows**. Further checks on categories and value ranges revealed several data-quality problems, summarised with the decision taken for each:

Issue found	Decision and justification
Redundant column 'Unnamed: 0' (old row index)	Dropped - it carries no information about the cars.
Inconsistent label: brand 'ISUZU' (2 rows) vs 'Isuzu' (8 rows)	Merged into 'Isuzu' so that one brand is not counted as two.
167 exact duplicate rows	Removed - identical listings would be double-counted in every statistic.
2 cars with seats = 0	Removed - a car cannot have zero seats, so these are invalid records.
2 cars with km_driven above 1,000,000 (3,800,000 km for a 5-year-old XUV500; 1,325,000 km for a 9-year-old E-Class)	Removed - physically implausible, most likely typing errors. Other high-km cars (up to 950,000 km) were kept as they are possible for taxis/commercial use.
Mileage (kmpl) given for 4 electric cars (all 19.16)	Set to NaN, not imputed - kmpl has no meaning for EVs. The 4 rows were kept.
Price outliers (e.g. Ferrari, Rolls-Royce, Bentley)	**Kept** - they are genuine listings, not errors. Removing them would hide the real price range. Handled with medians and log scales instead.
Data types	Text columns with few values (brand, seller_type, fuel_type, transmission_type) converted to 'category'. Two features added: price_lakh (price / 100,000) and km_per_year.

Dataset size before and after cleaning: 15,411 rows x 14 columns → 15,240 rows x 15 columns (171 rows removed, 1.1% of the data; the column count changes because the index column was dropped and two engineered columns were added).

Cleaning step	Rows remaining
Original dataset (incl. index column)	15,411
Dropped redundant index column 'Unnamed: 0'	15,411
Standardised brand label 'ISUZU' -> 'Isuzu'	15,411
Removed exact duplicate rows	15,244
Removed rows with 0 seats (invalid)	15,242
Removed km_driven > 1,000,000 (likely entry errors)	15,240
Set mileage of Electric cars to NaN (4 rows kept)	15,240
Added engineered columns price_lakh, km_per_year	15,240

Outlier detection (IQR rule: values beyond 1.5 x IQR from the quartiles). selling_price: 1,371 outliers (9.0%, above INR 14.85 lakh); km_driven: 462 (3.0%, above 130,000 km); engine: 2,101 (13.8%); max_power: 852 (5.6%); vehicle_age: 153 (1.0%, older than 14 years); mileage: 87 (0.6%). Most of these are luxury or large-engine cars, i.e. real observations, so only the clearly erroneous records described above were removed.

4. Descriptive Analysis

Numerical variables (cleaned data):

Variable	count	mean	std	min	25%	median	75%	max	skew
vehicle_age	15,240	6.04	3.01	0.00	4.00	6.00	8.00	29.00	0.83
km_driven	15,240	55,315	40,675	100.00	30,000	50,000	70,000	950,000	5.04
mileage	15,236	19.70	4.17	4.00	17.00	19.67	22.70	33.54	0.11
engine	15,240	1,486	520.32	793.00	1,197	1,248	1,582	6,592	1.67
max_power	15,240	100.60	42.91	38.40	74.00	88.50	117.30	626.00	2.49
seats	15,240	5.33	0.81	2.00	5.00	5.00	5.00	9.00	2.09
selling_price	15,240	774,653	894,753	40,000	385,000	559,000	825,000	39,500,000	10.11

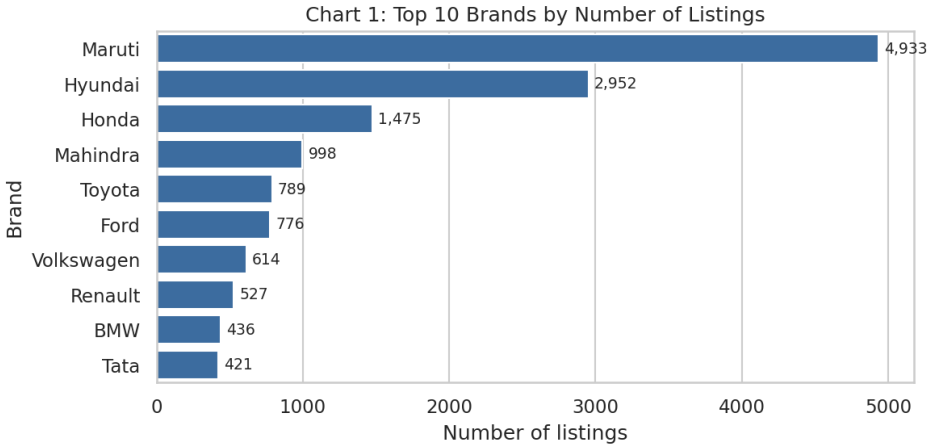
The typical car is 6 years old, has covered a median of 50,000 km, and costs a median of INR 5.59 lakh (mean 7.75 lakh). selling_price (skewness 10.1) and km_driven (skewness 5.0) are extremely right-skewed, so medians describe the 'typical' car better than means. Engine size and power are also right-skewed because of a small number of large luxury cars, while mileage is almost symmetric (skew 0.11).

Categorical variables (cleaned data):

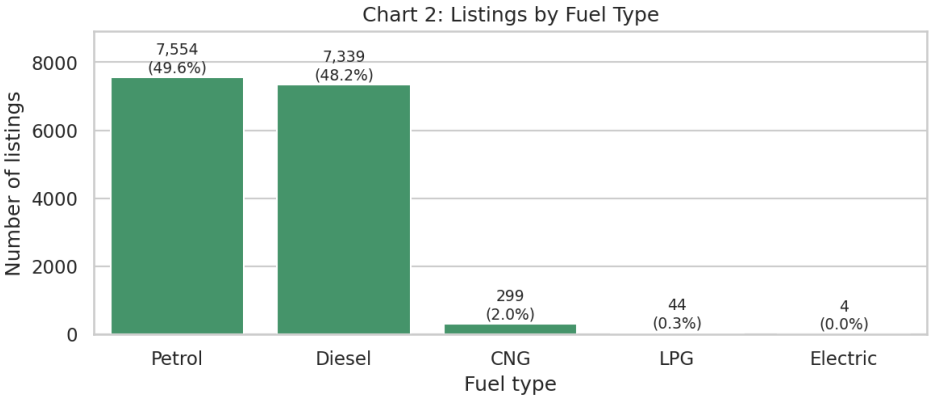
Variable	Unique values	Most common	Share of rows	Other values
brand	31	Maruti	32.4%	Hyundai 19.4%, Honda 9.7%, Mahindra 6.5%, Toyota 5.2%
fuel_type	5	Petrol	49.6%	Diesel 48.2%, CNG 2.0%, LPG 0.3%, Electric 4 cars
transmission_type	2	Manual	79.3%	Automatic 20.7%
seller_type	3	Dealer	62.1%	Individual 36.8%, Trustmark Dealer 1.1%

5. Visual EDA

Eight charts of five different types (bar, histogram, box plot, scatter plot and heatmap) cover univariate, bivariate and multivariate analysis. Because prices span from INR 0.4 lakh to 395 lakh, log scales are used where needed to keep the charts readable.

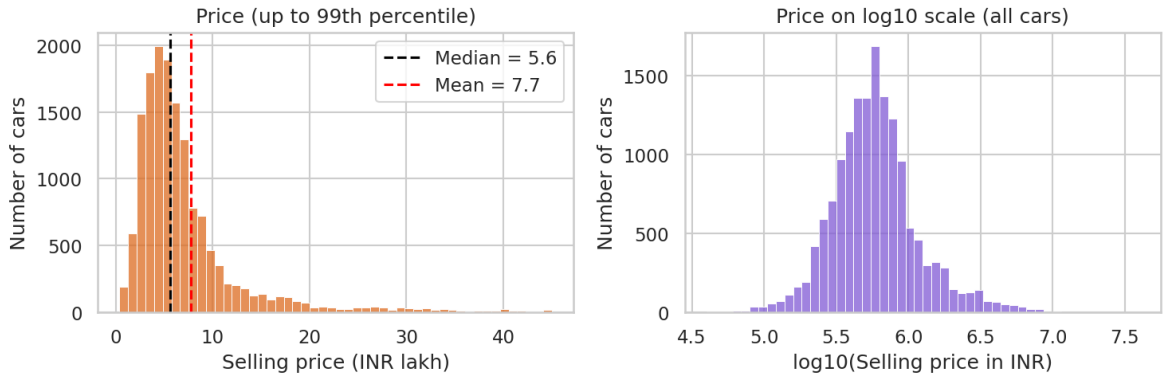


Observation: Maruti is by far the most listed brand with 4,933 cars, followed by Hyundai (2,952) and Honda (1,475). Maruti and Hyundai together make up 51.7% of all listings, so the used-car market in this data is dominated by mass-market brands.



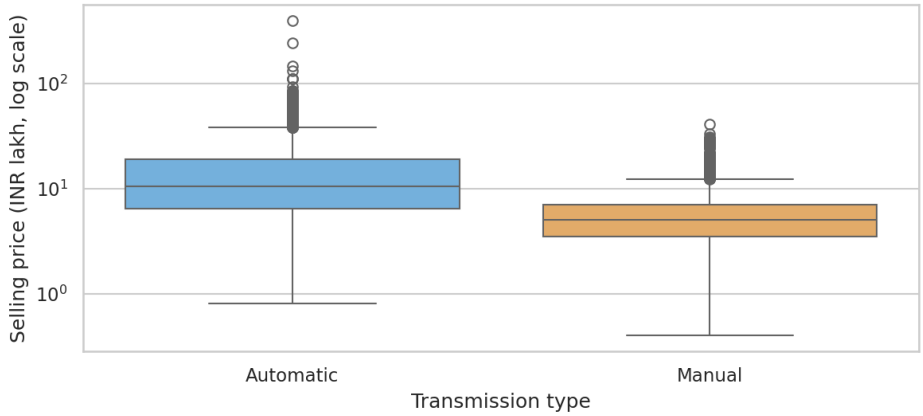
Observation: Petrol (7,554 cars, 49.6%) and diesel (7,339, 48.2%) are almost equally common and together account for 97.7% of listings. CNG (299), LPG (44) and Electric (4) are very rare, so any conclusion about the last two would be unreliable.

Chart 3: Distribution of Selling Price



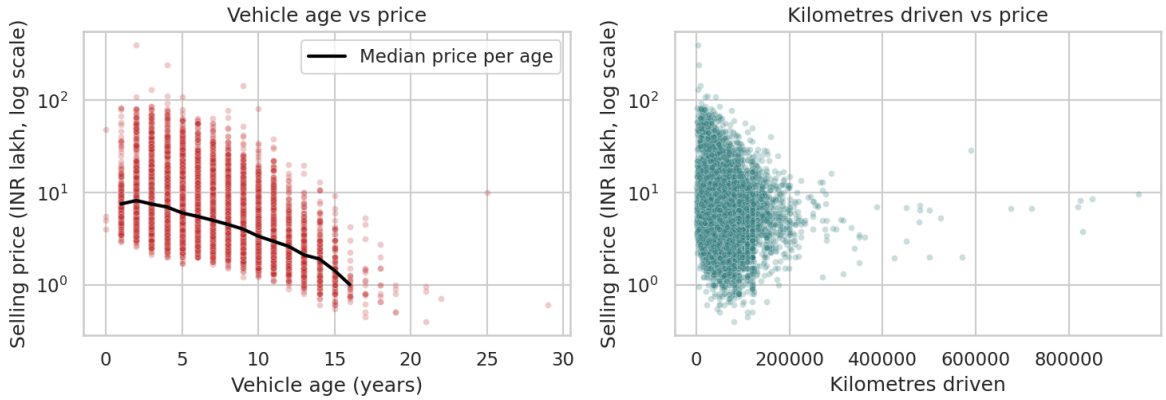
Observation: Price is strongly right-skewed: the median is 5.59 lakh but the mean is 7.75 lakh, because a few luxury cars (maximum 395 lakh) pull the average up. 99% of cars cost under 45.3 lakh. On the log scale (right) the distribution is close to a symmetric bell shape, so log-price is a better scale for analysis.

Chart 4: Selling Price by Transmission Type (log scale)

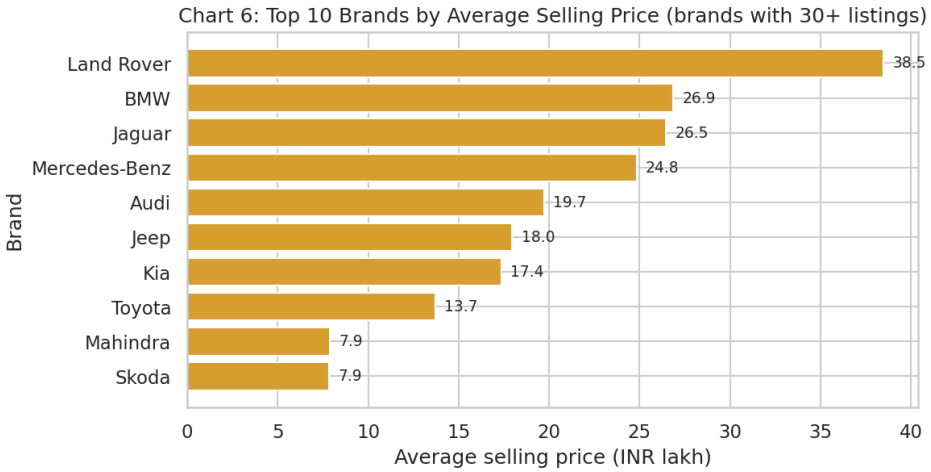


Observation: The median automatic car costs 10.5 lakh versus 5.0 lakh for a manual car (about 2.1 times higher), and the whole automatic box sits higher. Automatic cars are only 20.7% of listings and are concentrated in premium models, so this gap partly reflects the type of cars that have automatics, not the gearbox alone.

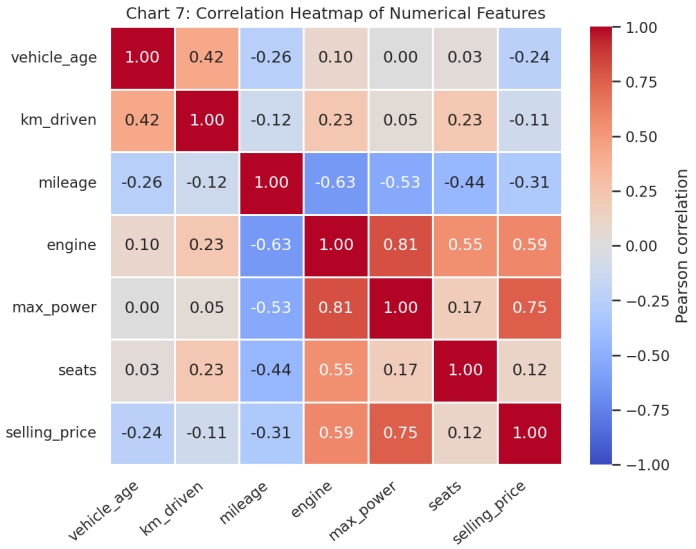
Chart 5: Does Age or Distance Driven Explain Price?



Observation: Price falls as cars get older: the median price is about 7.5 lakh for 1-3 year old cars, 4.5 lakh at 8 years, 3.4 lakh at 10 years and 1.4 lakh at 15 years (Spearman correlation -0.48). Kilometres driven shows a much weaker link with price (Spearman -0.17), partly because km and age move together (r = 0.42). The wide vertical spread at every age shows that brand and car segment matter a lot.

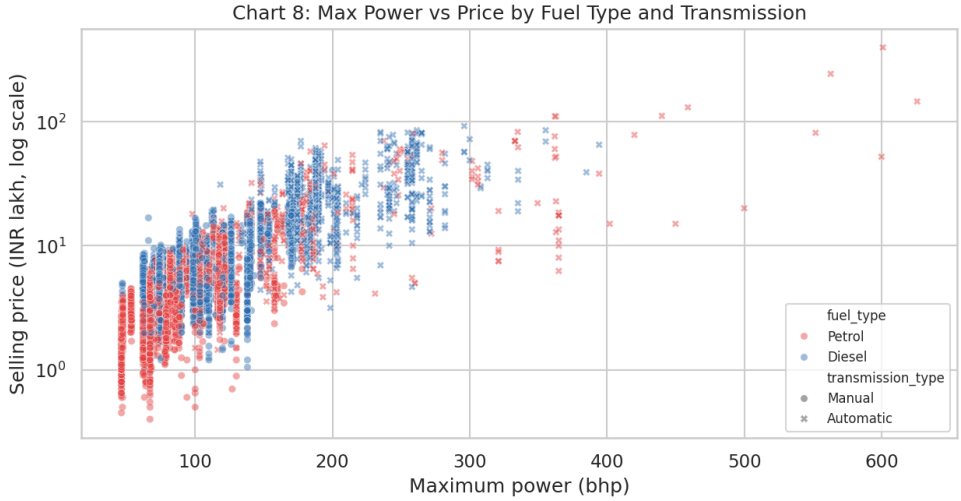


Observation: Among brands with at least 30 listings, Land Rover (38.5 lakh), BMW (26.9), Jaguar (26.5) and Mercedes-Benz (24.8) have the highest average prices, while volume brands such as Maruti and Hyundai do not appear in the top 10. Averages are lifted by a few very expensive cars (Mercedes-Benz: mean 24.8 vs median 18.0 lakh).



Observation: max_power has the strongest correlation with price (r = 0.75), followed by engine size (0.59). engine and max_power are themselves strongly correlated (0.81), and mileage falls as engine size rises (-0.63). Age (-0.24) and km_driven (-0.11) have weak linear correlations with price. These are associations, not proof of cause and effect.

Chart 8: Max Power vs Price by Fuel Type and Transmission



Observation: For petrol and diesel cars, price rises steadily with engine power (r = 0.75 on the raw price; the trend is roughly straight on the log axis, meaning price grows by a similar percentage per extra bhp). Automatic cars (cross markers) sit mostly in the upper part of the cloud. Petrol and diesel points overlap heavily at the same power, which suggests diesel's higher median price (7.0 vs 4.6 lakh) is largely explained by diesel being more common in bigger, more powerful cars.

6. Key Findings

- **The market is concentrated in a few mass-market brands.** Maruti and Hyundai account for 51.7% of the 15,240 cleaned listings, and the top 5 brands (adding Honda, Mahindra, Toyota) cover about 73% (Chart 1).
- **Petrol and diesel dominate.** Together they make up 97.7% of listings; CNG, LPG and Electric are marginal (Chart 2).
- **Prices are highly skewed.** Median 5.59 lakh vs mean 7.75 lakh (skewness 10.1), with 9% of cars flagged as price outliers (Chart 3). Medians should be preferred to means when describing price.
- **Automatic cars are priced about twice as high as manual cars** (median 10.5 vs 5.0 lakh), though part of the gap comes from automatics being concentrated in premium models (Chart 4, Chart 8).
- **Engine power is the feature most related to price** (r = 0.75), more than age (-0.24) or kilometres driven (-0.11) (Chart 7). Engine size and power overlap strongly (r = 0.81).
- **Age matters more than distance driven.** Median price drops from about 7.5 lakh (1-3 years) to 1.4 lakh (15 years); km_driven has only a weak relationship with price (Spearman -0.17) (Chart 5).
- **Luxury brands command the highest average prices**: Land Rover 38.5, BMW 26.9, Jaguar 26.5 and Mercedes-Benz 24.8 lakh (Chart 6).
- **Mileage is negatively related to price** (r = -0.31), most likely because powerful, expensive cars are less fuel-efficient (mileage vs max_power r = -0.53). This is an association, not a cause.

Limitations of the dataset

- The prices are **asking prices** from a single platform (CarDekho), not final transaction prices.
- There is no information on city, car condition, accident history, trim or listing date, so many real price drivers are missing.
- Only 4 electric and 44 LPG cars exist, so nothing reliable can be said about these fuel types.
- Correlation does not imply causation; features such as engine size and power are strongly related to each other.
- Brand-level averages for brands with few listings (e.g. Land Rover, 50 cars) are less stable than those for large brands.

7. Conclusion

Learning. The EDA showed that a seemingly clean dataset can still hide issues - duplicates, a repeated brand with different capitalisation, impossible seat counts and implausible odometer readings - and that each fix should be justified instead of applied automatically. Visualising price on a log scale and using medians gave a much more honest picture than raw means. The used-car prices in this dataset are driven mainly by engine power and brand segment, then by age; distance driven plays a smaller role.

Challenges. Deciding what is an error versus a genuine extreme (a Ferrari at 395 lakh is real, a 38-lakh-km reading is not), handling the heavy skew in price, and avoiding claims about causes that the data cannot support.

Future analysis. (1) Study price per segment (hatchback, sedan, SUV) after creating this feature from model names; (2) compare depreciation curves by brand; (3) analyse km_per_year to detect heavily used (commercial) cars; (4) if a model is later required, use log(price), and handle the strong engine-power correlation.

References

Dataset: CarDekho used-car listings (cardekho.com), accessed via GitHub repository manishkr1754/CarDekho_Used_Car_Price_Prediction (notebooks/data/cardekho_dataset.csv). Libraries: pandas, NumPy, Matplotlib, Seaborn. Currency: 1 lakh = INR 100,000.